# CALL CENTER ANALYSIS

**ARTIFACTS**

Microsoft Solutions Partner

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Data & AI
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Digital & App Innovation
Azure

**Submitted By**

**Tata Tele Business Service**

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# 1. Document objective

- The README for a project called **Call Center Analysis**, which analyses recorded customer calls on Azure: it transcribes each call, hides personal details, and scores the call for sentiment, quality and compliance.
- **How it's built:** One Azure template creates the whole environment. The size (tier) of every service is chosen in one settings file per environment. You deploy with one script command, or create the Azure services with the **Deploy to Azure** button and then install the application with the same script.
- **The application:**
  - A storage account receives the call recordings from the telephony system.
  - An automated pipeline (Azure Functions) runs each new recording through the steps below.
  - Azure AI Speech turns the recording into text and separates the agent from the customer.
  - Azure AI Language finds and hides personal details (names, phone and card numbers and similar) and measures the sentiment of each sentence.
  - Azure OpenAI writes a summary, finds the call reason, root cause and outcome, and scores the call against the quality checklist and compliance statements. It only receives text with personal details already hidden.
  - A web app on Azure App Service shows the results to supervisors, quality teams and agents, after they sign in with Microsoft Entra ID.
- **Supporting services:**
  - Cosmos DB holds one record per call for the web app and for a CRM system.
  - Analysis files in the data lake are the data source for Power BI reports.
  - Event Hubs (optional) sends a message to other systems when a call has been analysed.
  - Key Vault keeps the few secrets; Application Insights and Log Analytics provide logging and monitoring.
  - Managed identities let the applications connect to the Azure services without passwords or keys.
- **Intended use:** A ready-to-deploy starting point for automated quality review of contact-centre calls in English. The quality checklist and compliance statements included are samples, to be replaced with the customer's approved versions. It has been tested on a developer machine with simulated Azure services, but not yet deployed to a live Azure subscription, so deploy it to a test subscription first.
- **What the document covers:**
  - An overview of the solution, the terms used, and how it works
  - The screens and who can see what
  - How to try it without Azure
  - What you need before you start, and how to choose the size of each service
  - The ways to deploy it: the script, the button and GitHub Actions
  - What to do after deployment, including connecting the telephony system
  - How to use the web app, and cost, security and clean-up notes
  - Troubleshooting, the repository layout and the current status

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Call Center Analysis analyses every recorded customer call automatically. Each call is turned into text, personal details are hidden, and the call is scored for sentiment, emotion, quality (QA) and compliance. It also gets a summary, a call reason and a root cause. Supervisors, quality teams and agents see the results in a secure web app and in Power BI, and a CRM system can fetch them through an API or receive them as events.

The solution is a ready-to-deploy template. You choose the size (SKU) of every Azure service in one settings file, then deploy with one command or the **Deploy to Azure** button.

Deploy to Azure

The button creates the Azure services only. The application itself is then installed with `Deployment/scripts/deploy.sh` or the *Deploy* GitHub workflow (see "Deployment options"). The button only works while the GitHub repository is public.

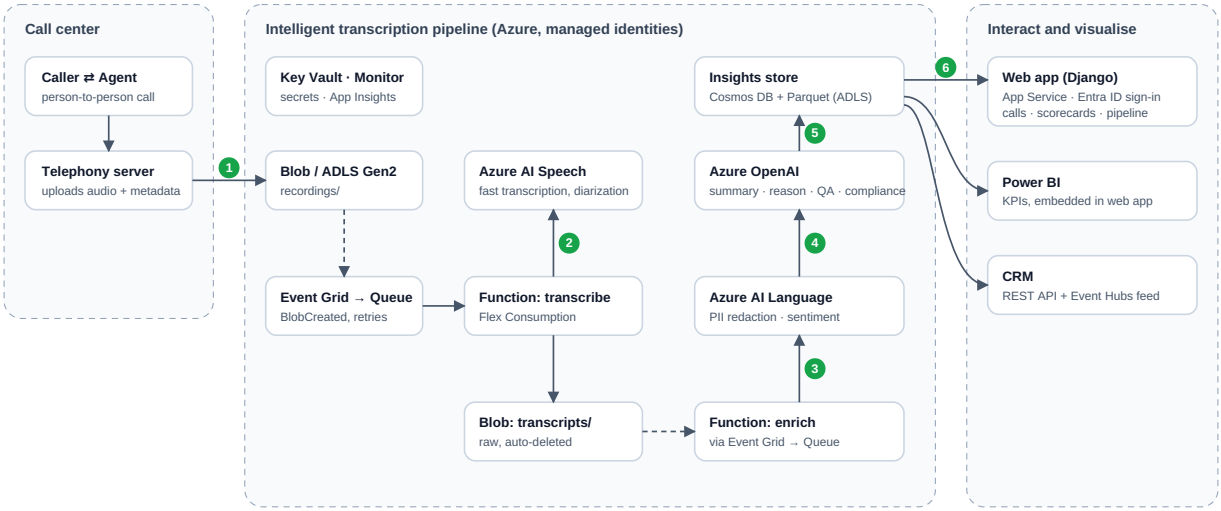
## Terms used in this document

| Term | Meaning |
|---|---|
| SKU | The size or pricing tier of an Azure service, for example S1 for a web app plan. |
| PII | Personally identifiable information: names, phone numbers, card and account numbers, email and postal addresses, dates of birth. |
| Redaction | Replacing personal details in the transcript with a label such as `[PHONE]` or `[CARD]`. |
| Diarization | Telling the speakers apart in a recording that has both voices on one channel. |
| QA scorecard | The list of quality criteria a call is scored against (for example greeting, verification, resolution), each worth a number of points. |
| Compliance check | A statement the agent must make (for example the call-recording notice) or something the agent must never do (for example ask for an OTP or CVV). |

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| Term | Meaning |
|---|---|
| LLM | Large language model, here Azure OpenAI, which reads the transcript and returns the summary and scores. |
| TPM | Tokens per minute: how much text Azure OpenAI will process each minute. A token is roughly three quarters of a word. Capacity is set in thousands. |
| Managed identity | An identity Azure gives a service so it can access other services without a stored password or key. |
| Queue / poison queue | A waiting list of work items. A call that keeps failing is moved to a separate "poison" list so it does not block others. |
| Parquet | A compact file format for tables, read by Power BI. |
| App role | A role (Admin, Supervisor, QA, Agent) assigned to a user in Microsoft Entra ID that decides what they can see. |
| Local mode | Running everything on one computer, with simulated Azure services, for trials and development. |

## Architecture




Numbers follow the TSD flow: 1 upload · 2 speech to text · 3 PII redaction and sentiment · 4 LLM insights · 5 store · 6 web app / Power BI / CRM

| # | Step | Azure service |
|---|---|---|
| 1 | The telephony server uploads the recording, and optionally a small file with call details | Blob Storage (data lake) |
| 2 | The recording is turned into text, with the agent and the customer told apart | Azure AI Speech |

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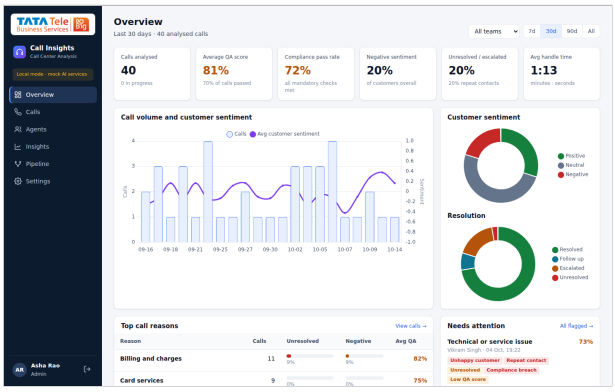
| # | Step | Azure service |
|---|---|---|
| 3 | Personal details are found and hidden, and the sentiment of each sentence is scored. This happens **before** any text reaches the language model. | Azure AI Language |
| 4 | Summary, call reason, root cause, outcome, customer emotion, QA scorecard and compliance checks | Azure OpenAI |
| 5 | Results are stored: one record per call for applications, and tables for reporting | Cosmos DB, data lake (Parquet) |
| 6 | Web app, Power BI report, CRM interface and event feed | App Service, Power BI, Event Hubs |

**How the work is passed along:** when a new file lands in storage, Azure Event Grid puts a message on a storage queue. A queue-triggered Azure Function (on the serverless *Flex Consumption* plan) picks it up. The two stages, transcription and analysis, each have their own queue. A failing message is retried up to 5 times and then moved to a poison queue; the call then shows as *failed* on the Pipeline page, with a **Retry** button.

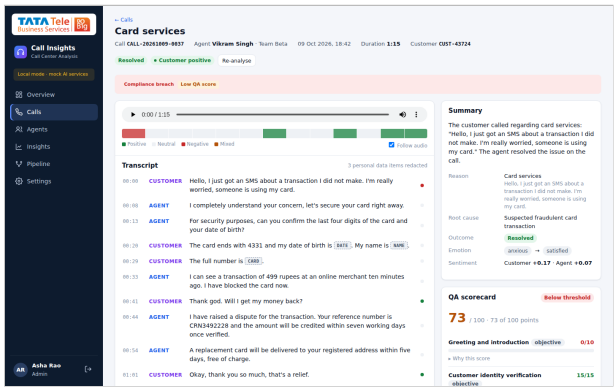
The web app and the Functions connect to the Azure services with **managed identities**. The few secrets (the web app's signing key, the sign-in client secret and the CRM API key) are kept in Key Vault.

More detail: Architecture · Technical specification

## Screens




*Overview*




*Call detail*

The screenshots were taken in local mode with synthetic sample calls (see "Trying it without Azure").

- **Overview:** headline figures, call volume and sentiment over time, top call reasons, calls that need attention, root causes and an agent leaderboard.
- **Calls:** search and filter by agent, team, reason, sentiment, outcome, flag and QA score range.
- **Call detail:** the recording with the redacted transcript beside it (click a line to jump to it), a sentiment strip, the summary, the QA scorecard with quotes as evidence, the compliance checklist and coaching tips.


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- **Agents / My scorecard:** QA, compliance and sentiment for each agent, and the criteria to coach on.
- **Insights:** the Power BI report, embedded. Until a report is connected, the page shows built-in charts.
- **Pipeline:** processing status, failed calls with a retry button, and a form to upload a test recording.
- **Settings:** edit the QA scorecard, compliance checks and call reasons. No redeployment is needed.

Who sees what is controlled by app roles in Microsoft Entra ID:

| Role | Access |
|---|---|
| **Admin** | Everything, including Settings |
| **Supervisor** | All calls, agents, insights and the Pipeline page (not Settings) |
| **QA** | All calls, agents and insights |
| **Agent** | Their own calls and their own scorecard |

## Trying it without Azure

Local mode runs the whole solution on one computer. It replaces each Azure service with a local stand-in: a folder instead of Storage, a SQLite file instead of Cosmos DB, simulated Speech, Language and OpenAI services that return realistic results, and demo users instead of Entra ID sign-in. It is meant for trying the screens and for development. On the sign-in page, pick a demo user (Admin, Supervisor, QA or an agent):

```
make setup     # creates .venv and installs everything
make seed      # generates and analyses 40 synthetic English calls
make run       # opens the web app at http://localhost:8000
make test      # runs the automated tests
```

Without `make`:

```
python3 -m venv .venv && . .venv/bin/activate
pip install -r requirements-dev.txt
python -m callcenter seed
cd Code/webapp && python manage.py runserver
```

**Moving to real services one at a time:** copy `.env.example` to `.env` and set the variables for one service, for example `LLM_BACKEND=azure` with `AZURE_OPENAI_ENDPOINT`, `AZURE_OPENAI_KEY` and `AZURE_OPENAI_DEPLOYMENT`. Everything else stays simulated. The `make` commands read `.env`; when you run Python directly, set the variables in your shell instead.

`make docker-up` is an alternative that runs the web app in Docker with the Azurite storage emulator.

More detail: Local development · Configuration

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## Prerequisites

| Requirement | Notes |
|---|---|
| Azure subscription | **Owner** on the resource group, or **Contributor** plus **User Access Administrator**, because the template gives the services access to each other. |
| Microsoft Entra ID | Permission to create app registrations (members have it by default). Assigning roles to other people needs an administrator role such as Cloud Application Administrator. |
| Azure OpenAI quota | Access and quota for the chosen model and deployment type in the Azure OpenAI region. Check it in *Azure AI Foundry → Quotas*. |
| Resource providers | `Microsoft.Web`, `Microsoft.CognitiveServices`, `Microsoft.DocumentDB`, `Microsoft.EventGrid`, `Microsoft.EventHub`, `Microsoft.KeyVault`, `Microsoft.OperationalInsights`, `Microsoft.Insights` and `Microsoft.Storage` must be registered. Register a missing one with `az provider register -n <name>`. |
| Tools | Azure CLI (2.60 or later recommended), bash, `rsync`, `zip` and `python3`. On Windows, use WSL or Azure Cloud Shell; the scripts are bash only. |

## Choosing SKUs

All service sizes are kept in settings files in `Deployment/ARM/parameters/`, one per environment. Two are included:

| File | Meant for | Main choices |
|---|---|---|
| `dev.parameters.json` | Trials and development | Web app B1; Functions up to 40 instances; Azure OpenAI 30K TPM; Cosmos DB serverless; Event Hubs off; recordings kept 90 days |
| `prod.parameters.json` | Production | Web app S1; Functions up to 40 instances; Azure OpenAI 100K TPM; Cosmos DB serverless; Event Hubs on; recordings kept 365 days |

To add an environment, copy a file (for example to `uat.parameters.json`) and deploy with `-e uat`.

What each setting controls:

| Setting | What it affects | Allowed values |
|---|---|---|
| `location` | Region for every service except Azure OpenAI (default `centralindia`) | Any region that offers the services |
| `storageSku` | How many copies of the recordings Azure keeps, and where | `Standard_LRS`, `Standard_ZRS`, `Standard_GRS`, `Standard_RAGRS`, `Standard_GZRS` |
| `recordingCoolAfterDays`, `recordingDeleteAfterDays` | When recordings move to cheaper storage, and when they are deleted | Days; delete must be later than the move |

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| Setting | What it affects | Allowed values |
|---|---|---|
| `rawTranscriptDeleteAfterDays` | When the original transcripts, which still contain personal details, are deleted | Days (the redacted results are kept) |
| `speechSku`, `languageSku` | Azure AI Speech and Language pricing tier | `S0` or `F0`; `S` or `F0` (free tiers are limited and allowed once per subscription) |
| `openAiLocation` | Where the transcript text is processed by Azure OpenAI, and which models are available | A region that offers the model |
| `openAiModelName`, `openAiModelVersion` | Quality, speed and price of the analysis (default `gpt-4.1-mini`, `2025-04-14`) | A chat model that supports structured (JSON) output |
| `openAiDeploymentSku` | Where data is processed, and how much quota is available by default | `Standard` (within the region), `DataZoneStandard`, `GlobalStandard` |
| `openAiCapacity` | The most text processed per minute, in thousands of tokens. Must fit your quota. | 1 or more |
| `cosmosCapacityMode`, `cosmosThroughput` | Database billing: per request, or a fixed reserved throughput | `Serverless`, `Provisioned` (400 RU/s or more) |
| `webAppSku`, `webAppInstanceCount` | Web app cost and capacity | `B1`-`B3`, `S1`-`S3`, `P0v3`-`P3v3`; 1 to 10 instances |
| `functionMaximumInstances`, `functionInstanceMemoryMB` | How many calls are processed in parallel (this also limits the load on Speech, Language and OpenAI), and memory per instance | 40 to 1000 (billed per execution, not per instance); `512`, `2048` or `4096` MB |
| `deployEventHubs`, `eventHubSku`, `eventHubCapacity` | Whether the event feed is created, and its tier and throughput units | `true`/`false`; `Basic`, `Standard`; 1 to 20 |
| `logAnalyticsRetentionDays`, `logAnalyticsDailyQuotaGb` | How long logs are kept, and the most log data accepted per day | 30 to 730 days; GB per day, or -1 for no limit |
| `keyVaultSku` | Key Vault tier | `standard`, `premium` |
| `entraDefaultRole` | Role for users who sign in without an assigned role | `Agent`, `QA`, `Supervisor`, or empty to refuse access |

The Deploy to Azure button shows the same choices as drop-down lists and sliders. The full list, with the value in each file, is in the SKU reference.

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## Deployment options

### 1. Script (recommended)

```
az login
az account set --subscription "<subscription>"
Deployment/scripts/deploy.sh -g rg-callcenter-dev -e dev --setup-entra
```

The script:

1. Creates the resource group (in `centralindia` unless you pass `-l`).
2. Deploys the template with your settings file, and gives you access to the storage account and Key Vault for the steps that follow.
3. Packages the application and installs it on the Function app and the web app. Azure installs the Python libraries during this step.
4. With `--setup-entra`: creates the Entra ID app registration with the four app roles, stores its client secret in Key Vault, configures the web app, writes the client ID into your settings file, and gives you the **Admin** role.

| Option | What it does |
|---|---|
| `-g`, `--resource-group` | Resource group (created if it does not exist). Required. |
| `-e`, `--environment` | Which settings file to use (`dev`, `prod`, ...). Required. |
| `-l`, `--location` | Region for the resource group (default `centralindia`) |
| `-s`, `--subscription` | Azure subscription to use |
| `--setup-entra` | Create or update the sign-in app registration and roles |
| `--skip-infra` | Install only the application code |
| `--skip-code` | Create only the Azure services |
| `--seed` | Upload 20 synthetic sample calls as a test. Needs the local Python environment from `make setup`, activated. |

You can safely run the script again: it updates what is already there.

### 2. Deploy to Azure button

The button opens a form in the Azure portal with tabs for the AI services, storage and database, compute and messaging, and sign-in and reporting. It creates the Azure services only. Afterwards, install the application and set up sign-in:

```
Deployment/scripts/deploy.sh -g <resource-group> -e <environment> --skip-infra --setup-entra
```

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If you created the app registration yourself, enter its client ID and secret on the form's sign-in tab instead. Use the redirect address `https://<web app>/auth/callback` and create app roles with the values `Admin`, `Supervisor`, `QA` and `Agent`.

### 3. GitHub Actions

The repository includes a workflow that deploys a chosen environment from GitHub without storing any passwords. It needs a one-time setup, described in the Deployment guide. Run the sign-in setup (`setup_entra.sh`) once from your own computer, because GitHub's identity normally cannot create app registrations.

## After deployment

1. **Sign in** at the web app address printed at the end of the script. If the page says sign-in is not configured, run `Deployment/scripts/setup_entra.sh -g <resource-group> -e <environment>`.
2. **Give people access:** in the Entra admin center, open *Enterprise applications → Call Center Analysis (<env>) → Users and groups → Add assignment*. Assigning groups (for example the QA team to the QA role) is easiest. People without a role get the `entraDefaultRole` (Agent by default), so they see only calls recorded under their own email address.
3. **Test the pipeline:** on the Pipeline page, upload an English call recording and watch it move from *transcribing* to *completed*.
4. **Connect the telephony system** (see the next section).
5. **Replace the samples:** on the Settings page, replace the sample QA scorecard, compliance checks and call reasons with the customer's approved versions.
6. **Connect the CRM:** read the API key with

   ```
   az keyvault secret show --vault-name <key-vault> -n crm-api-keys --query value -o tsv
   ```

   and call `GET https://<web app>/api/calls` with the header `X-API-Key: <key>`. To receive events instead, give the CRM integration the *Azure Event Hubs Data Receiver* role on the `call-insights` event hub (only when `deployEventHubs` is `true`).
7. **Power BI (optional):** follow Analytics/README.md. Embedding the report in the web app needs a Power BI Embedded or Fabric capacity, which this template does not create.
8. **Defender for Cloud:** this is set for the whole subscription and is not part of the template. Turn on the plans you need in *Defender for Cloud → Environment settings*.

Two things to know when you deploy again:

- Each deployment of the Azure services creates a **new CRM API key** and a new web app signing key, so users sign in again and the CRM must be given the new key. To keep the CRM key, pass `crmApiKey` as a parameter.
- The sign-in client secret created by `setup_entra.sh` is valid for **one year**. Run the script again before it expires.

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Full steps and troubleshooting: Deployment guide

## Connecting the telephony system

| Item | What to do |
|---|---|
| Where | Container `recordings` in the storage account created by the template. Folders are allowed. |
| Audio files | `.wav` (preferred), `.mp3`, `.ogg`, `.opus`, `.flac`, `.m4a`, `.webm`, `.wma` or `.aac`, up to 2 hours each. |
| Call ID | The file name without its extension, which must be unique, for example `CALL-123456.wav`. |
| Call details (optional) | A JSON file with the same name (`CALL-123456.json`), uploaded **before** the audio, with fields such as `agent_id`, `agent_name`, `agent_email`, `team`, `customer_id`, `started_at` and `channels`. |
| Access | The *Storage Blob Data Contributor* role on the container, or a write-only access token limited to `recordings`. |

`agent_email` must match the agent's sign-in address so they see their own calls. Stereo recordings (`"channels": 2`, agent on channel 0 by default) give the most reliable split between agent and customer. The field list is in the Technical specification.

## Using the web app

- **Date range:** most pages have *7d*, *30d*, *90d* and *All* buttons. The default is the last 30 days.
- **Calls:** type in the search box or change a filter and the list updates straight away. Select a column title to sort. The list shows 25 calls per page.
- **Call detail:** select a line of the transcript, or a point on the sentiment strip, to play the recording from there. *Follow audio* keeps the current line in view. Personal details appear as grey labels such as `PHONE`. Under each QA criterion, *Why this score* shows the evidence and the reasoning.
- **Re-analyse / Retry** (Admin and Supervisor): runs a call through the pipeline again, for example after the scorecard was changed. Saving new settings does not change calls that were already analysed.
- **Pipeline page:** refreshes every 5 seconds. The upload form accepts the audio types listed above, up to 200 MB, plus optional call details and a *Stereo recording* tick box.
- **Settings** (Admin): *Validate and save* checks the file and refuses it with an explanation if it is not valid. *Reset to default* restores the version from the repository.
- **CRM interface:** `GET /api/calls` (filters: `from`, `to`, `agent`, `team`, `reason`, `status`, `flag`, `customer`, `limit`, up to 1000) and `GET /api/calls/<call ID>` for one call, with its redacted transcript.

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## Cost, security and clean-up

### Cost notes

- **Charged all the time, even with no calls:** the web app plan (per instance), Event Hubs (per throughput unit, when deployed) and Cosmos DB in *Provisioned* mode. The Functions, Speech, Language and OpenAI are charged only when used.
- **Charged by use:** Azure AI Speech (by hours of audio), Azure AI Language (by amount of text), Azure OpenAI (by tokens), Cosmos DB in *Serverless* mode (by requests), storage (by size), and log data (limited by `logAnalyticsDailyQuotaGb`).
- **Not in the template:** Power BI licences or capacity, and Microsoft Defender for Cloud plans.
- Check current prices for your region before deploying: https://azure.microsoft.com/pricing/calculator/

### Security notes

- **Sign-in and roles:** users sign in with Microsoft Entra ID, and app roles decide what they can see. Agents see only their own calls.
- **No keys in the application settings:** the Functions and the web app use managed identities, each with only the access it needs. Key-based access is still switched on in Storage, Cosmos DB and the AI services (so Power BI and local tools can connect) but the applications do not use it; it can be switched off.
- **Secrets in Key Vault:** the web app's signing key, the sign-in client secret and the CRM API key.
- **Personal data:** transcripts are redacted **before** Azure OpenAI is called. The screens, Cosmos DB, the Power BI tables and the CRM interface only receive redacted text. The original transcripts are deleted automatically after `rawTranscriptDeleteAfterDays`.
- **Connections:** HTTPS only, TLS 1.2 or later, secure cookies and protection against cross-site request forgery. The services are reachable over the internet; private network connections and a web application firewall are not part of this template.
- More detail: SECURITY.md

### Clean-up

```
az group delete --name rg-callcenter-dev --yes --no-wait
```

Azure keeps deleted Key Vaults and AI service accounts for a while (soft delete). To deploy again into a resource group with the same name and the same `solutionPrefix`, remove them permanently first:

```
az keyvault purge --name <key-vault-name>
az cognitiveservices account purge --name <account-name> --resource-group rg-callcenter-dev
--location <region>
```

Run the second command for each of the Speech, Language and Azure OpenAI accounts. Also delete the Entra ID app registration *Call Center Analysis (<env>)* if it is no longer needed.

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## Troubleshooting

| What you see | What to do |
|---|---|
| Deployment fails while creating the Azure OpenAI model | The model, version or deployment type is not offered in `openAiLocation`, or there is not enough quota. Change the settings or ask Azure for more quota. |
| `AuthorizationFailed` while access is being set up | You need Owner, or Contributor plus User Access Administrator, on the resource group. |
| Calls stay *uploaded* or *transcribing* | Check the Function app's log stream or Application Insights, that the Event Grid subscriptions show *Succeeded*, and that the `recordings-events` queue receives messages. |
| `403` errors from Speech, Language or OpenAI in the logs | New access permissions can take several minutes to apply after the first deployment. Retry the call from the Pipeline page. |
| A call fails at the *transcribe* stage | The audio format is not supported or the file is empty. Convert the recording to WAV. |
| The web app shows "Application Error" | Look at the App Service log stream. A Key Vault reference may not be readable yet; restart the web app once permissions apply. |
| Sign-in keeps returning, or error `AADSTS50011` | The redirect address must be exactly `https://<web app>/auth/callback`. Run `setup_entra.sh` again. |
| A user signs in but sees no calls | Agents see only calls whose `agent_email` matches their sign-in email. Give supervisors the Supervisor or QA role. |

More: Deployment guide → Troubleshooting

## Repository layout

| Folder | Contents |
|---|---|
| `Code/core` | The processing pipeline, the connections to Speech, Language and OpenAI (real and simulated), storage, reporting figures and command-line tools |
| `Code/functions` | The Azure Functions app that runs the two pipeline stages and handles calls that keep failing |
| `Code/webapp` | The web app: screens, Entra ID sign-in, CRM interface and Power BI embedding |
| `Config` | QA scorecard, compliance checks and call reasons (editable at runtime) |
| `Deployment/ARM` | The template, the settings files per environment, and the portal form for the button |
| `Deployment/scripts` | `deploy.sh`, `setup_entra.sh` and `package.sh` |
| `Analytics` | Power BI data model: data sources, measures, report and embedding guide |
| `Data/sample_calls` | Synthetic call scripts for local mode and demonstrations |

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| Folder | Contents |
|---|---|
| `Documents` | Architecture, technical specification, deployment guide, configuration and local development |
| `.github/workflows` | Automatic checks on every change, and the manual deployment workflow |

## Current status

| Item | Status |
|---|---|
| Post-call analysis: recording, transcription, hiding personal details, sentiment, QA, compliance, summaries, web app, Power BI data, CRM interface | Built; tested on a developer machine with simulated Azure services and the Azurite storage emulator. The QA scorecard and compliance checks are samples. |
| First deployment to a live Azure subscription (including model availability in the default region) | Still to do: needs a subscription with Azure OpenAI quota |
| Real-time transcription and suggestions to agents during the call | Planned. Event Hubs is created when `deployEventHubs` is `true` (the prod settings file). |
| Languages other than English | Not yet: needs code changes as well as settings, plus accuracy testing |

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